

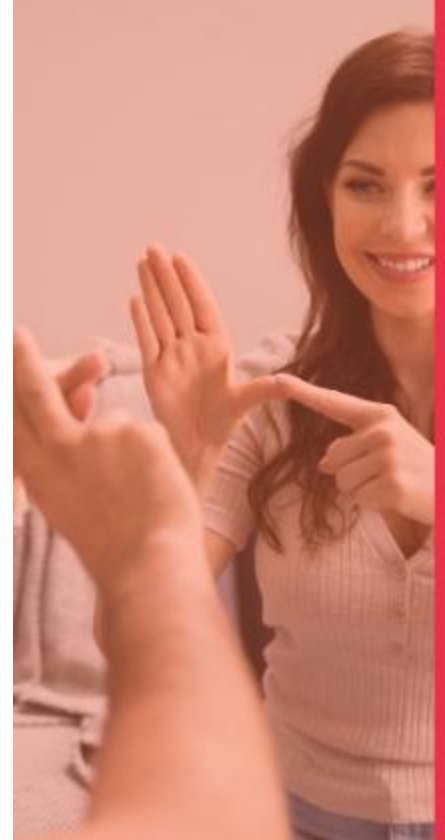
SIGN LANGUAGE DETECTION USING YOLOV5

Real-time gesture recognition — Implementation & Project Overview

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PROBLEM STATEMENT

- Communication barrier for hearing and speech impaired individuals.
- Lack of scalable, real-time, automated sign language translation systems.
- Need for accessible tools to convert gestures to text or speech.



PROJECT OBJECTIVES

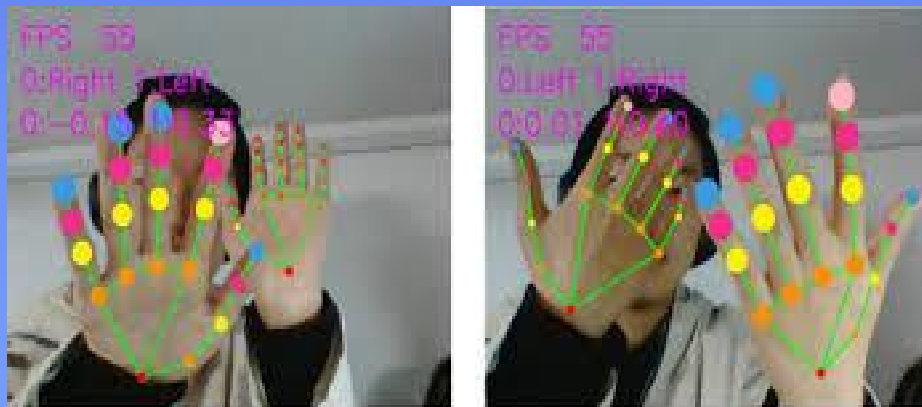
- Develop real-time sign detection using YOLOv5.
- Translate detected gestures into text and speech.
- Improve robustness via augmentation and multilingual support.
- Deployable on desktop and edge devices.



**SIGN LANGUAGE
RECOGNITION SYSTEM**

YOLO V5

- You Only Look Once v5 — real-time object detection model (PyTorch).
- Detects and classifies objects in a single pass for speed.
- Lightweight, accurate, and suitable for custom gesture datasets.



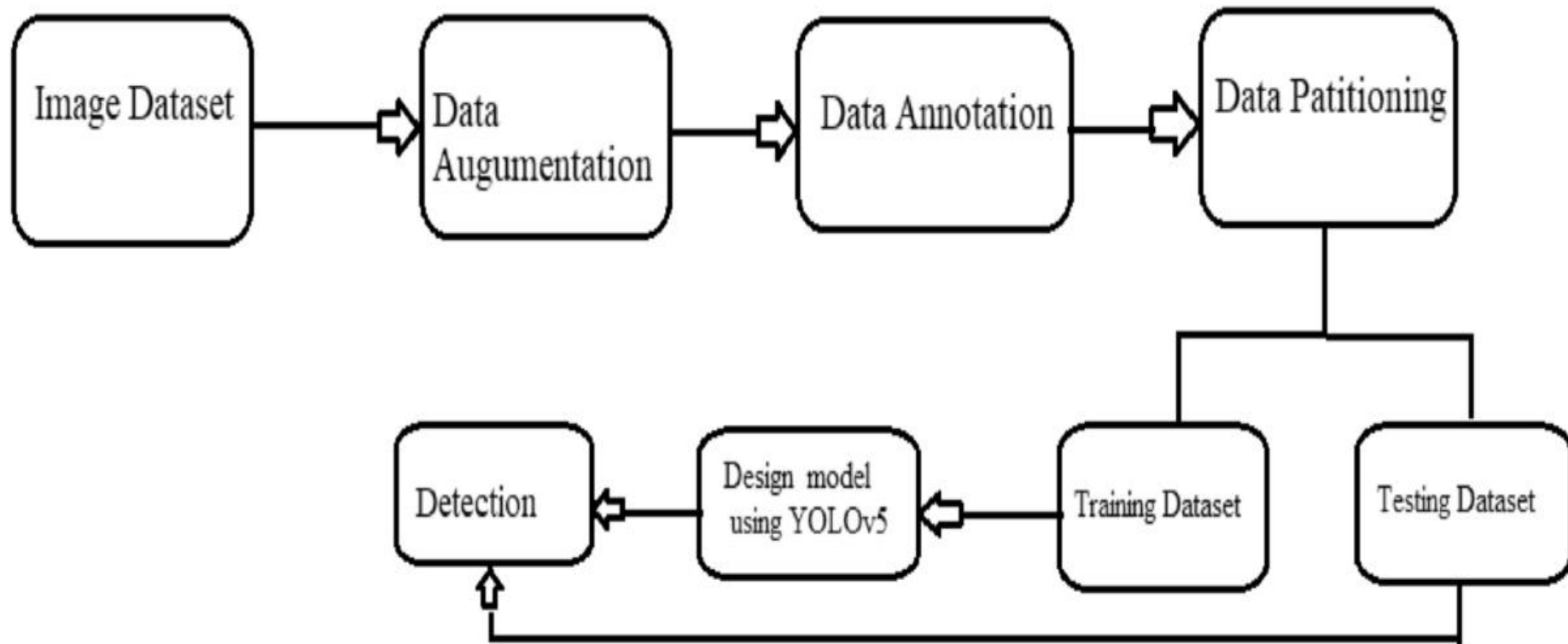
(a)

(b)

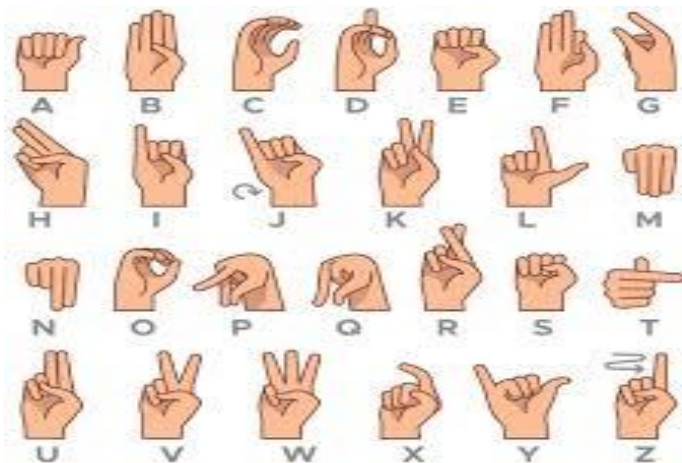


SYSTEM ARCHITECTURE (FLOW)

Video Input → Preprocessing (resize, normalization, augmentation) → YOLOv5 Detection → Post-processing (NMS, smoothing) → Translation (text/TTS) → UI/Output



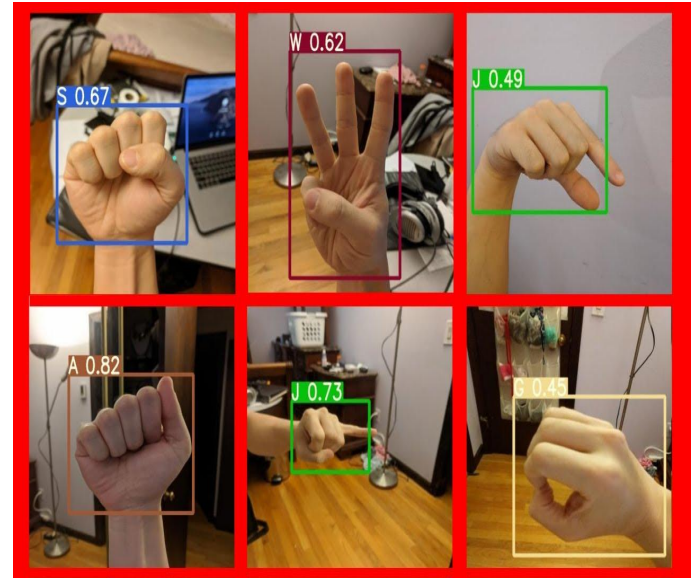
DATASET PREPARATION



- Collect images/videos for chosen sign language (ASL/ISL).
- Annotate bounding boxes using Labellmg (classes = sign labels).
- Apply augmentations: rotation, flip, color jitter, scaling.
- Split: train / val / test (e.g., 70 / 20 / 10).

MODEL TRAINING

- **Use YOLOv5 (Ultralytics)** implementation in PyTorch.
- **Transfer learning:** fine-tune a pre-trained YOLOv5s model.
- **Sample hyperparameters:** epochs=100, batch=16, img=640, lr=0.01.
- Use early stopping and monitor mAP, precision, recall.

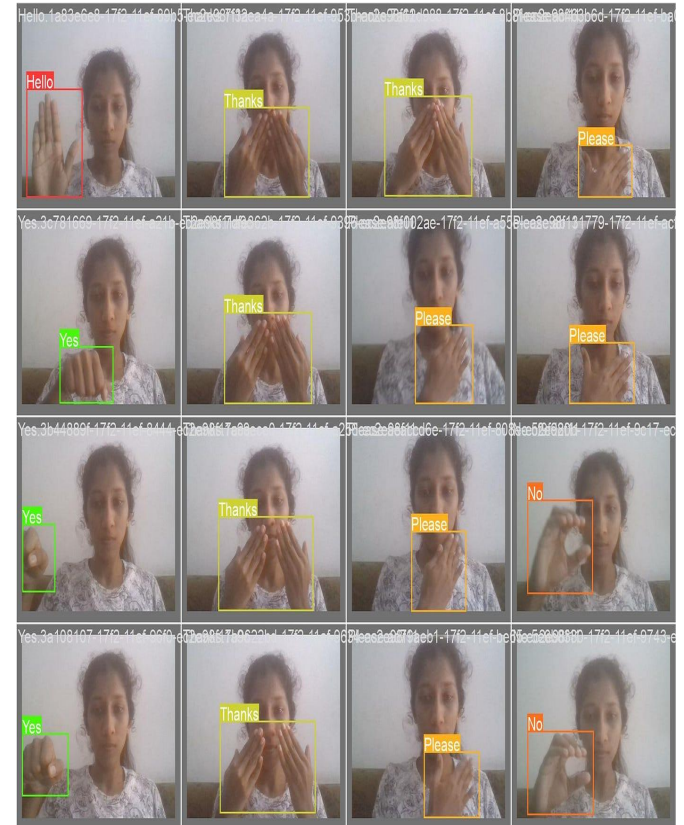


SYSTEM DEVELOPMENT & DEPLOYMENT

- Inference pipeline using OpenCV (video capture & preprocessing).
- Load trained model with PyTorch / TorchScript / ONNX for speed.
- Text-to-Speech: pyttsx3 or gTTS for voice output.
- Deploy options: Flask web app, mobile (ONNX/TFLite), Raspberry Pi.

Testing & Evaluation

- Evaluate on test set:
mAP@0.5, precision, recall,
FPS.
- User testing in varied lighting,
backgrounds, and signers.
- Measure latency for real-time
suitability (target 15+ FPS).
- Iterate with more data and
augmentations to reduce
errors.



INNOVATIONS & FUTURE WORK

- Continuous gesture-to-text (buffered sequence recognition).
- Multilingual translation and mobile/edge deployment.
- AR overlay, emotion + sign fusion, user-customizable signs.
- Feedback system for gesture correction and learning mode.





CONCLUSION

- YOLOv5 enables practical, real-time sign language detection.
- Combining detection with TTS and translation improves accessibility.
- Project demonstrates social impact and potential for deployment.



REFERENCES

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- [2] M. Shah and S. Jain, "Real-time Indian Sign Language Detection Using YOLOv3 and CNN," in *Proc. Int. Conf. Intell. Technol. (CONIT)*, IEEE, pp. 1–6, Jun. 2021.
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